

uniform Approximation Theory (UAT)

if f is a continuous function on a compact set such as K , then for any desired accuracy $\varepsilon > 0$, one can construct a feed-forward neural-network with one hidden layer and enough sigmoidal neurons, such that the network's output $N(x)$ differs from $f(x)$ by at most ε at every point of K .

Theorem

if f is a continuous function on a compact set K , then for every $\varepsilon > 0$, there exists a feed-forward network with a single hidden layer and sufficient number of sigmoidal neurons such that the network's output is at most ε away from f .

that is

$$\forall f \in C(K), \forall \varepsilon > 0, \exists N$$

such that

$$\sup_{x \in K} |f(x) - N(x)| < \varepsilon$$